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Smoked drug use in patients with TB is associated with higher bacterial burden

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SUMMARY

BACKGROUND: Smoking of illicit drugs may lead to more rapid TB disease progression or late treatment presentation, yet research on this topic is scant. We examined the association between smoked drug use and bacterial burden among patients newly initiated on drug-susceptible TB (DS-TB) therapy.

METHODS: Data from 303 participants initiating DS-TB treatment in the Western Cape Province, South Africa, were analyzed. Smoked drug use was defined as self-reported or

biologically verified methamphetamine, methaqualone and/or cannabis use. Proportional hazard and logistic regression models (adjusted for age, sex, HIV status and tobacco use) examined associations between smoked drug use and mycobacterial time to culture positivity (TTP), acid-fast bacilli sputum smear positivity and lung cavitation.

RESULTS: People who smoked drugs (PWSD) comprised 54.8% ($n = 166$) of the cohort. TTP was faster for PWSD (hazard ratio 1.48, 95% CI 1.10–1.97; $P = 0.008$). Smear positivity was higher among PWSD (OR 2.28, 95% CI 1.22–4.34; $P = 0.011$). Smoked drug use (OR 1.08, 95% CI 0.62–1.87; $P = 0.799$) was not associated with increased cavitation.

CONCLUSIONS: PWSD had a higher bacterial burden at diagnosis than those who do not smoke drugs. Screening for TB among PWSD in the community may facilitate earlier linkage to TB treatment and reduce community transmission.

RÉSUMÉ

La consommation de stupéfiants peut augmenter le risque de présentation tardive pour traitement antituberculeux, mais les recherches sur ce sujet sont rares. Nous avons examiné l'association entre la consommation de drogues fumées et la charge bactérienne chez les patients nouvellement placés sous traitement pour TB pharmacosensible (DS-TB).

Les données de 303 participants commençant un traitement contre la DS-TB dans la province du Cap occidental, Afrique du Sud, ont été analysées. La consommation de drogues fumées a été définie comme la consommation de méthamphétamine, de méthaqualone et/ou de cannabis, auto-déclarée ou vérifiée biologiquement. Des modèles de hasard proportionnel et de régression logistique (ajustés pour l'âge, le sexe, le statut VIH et le tabagisme) ont permis d'examiner les associations entre la consommation de drogues fumées et le délai avant positivité des cultures mycobactériennes (TTP), la positivité des frottis d'expectoration aux bacilles acido-alcoolo-résistants et la cavitation pulmonaire.

Les fumeurs de drogues (PWSD) représentaient 54,8% ($n = 166$) de la cohorte. Le TTP était plus rapide chez les PWSD (rapport de risque 1,48 ; IC 95% 1,10–1,97 ; $P = 0,008$). La positivité des frottis était plus élevée chez les PWSD (OR 2,28 ; IC 95% 1,22–4,34 ; $P = 0,011$). La consommation de drogues fumées (OR 1,08 ; IC 95% 0,62–1,87 ; $P = 0,799$) n'était pas associée à une augmentation de la cavitation.

La charge bactérienne au moment du diagnostic était plus élevée chez les PWSD que chez les non-fumeurs de drogues. Le dépistage communautaire de la TB chez les PWSD peut permettre la prescription précoce du traitement antituberculeux et réduire la transmission au sein de la communauté.

Keywords

smoked drug use; drug-susceptible tuberculosis; South Africa; bacterial load

Despite effective treatment regimens, TB is the leading infectious disease killer globally.¹ Studies indicate a high prevalence of TB among people who use drugs, with people who used alcohol or drugs consistently overrepresented in TB outbreaks in the United States between 2009 and 2015.² TB outbreaks among people who use drugs have been reported

in Argentina, Canada, England, Latvia, Portugal and Thailand.^{3–8} Similar patterns are likely to occur in high TB prevalence settings (like South Africa) but may be masked by TB endemicity.

Physiologic effects of illicit drugs along with environmental and behavioral risks may contribute to the high TB prevalence among people who use drugs.³ Illicit drug use may increase host susceptibility to mycobacterial infection by impairing cell-mediated immune response.^{9,10} Furthermore, illicit drug use is frequently associated with other risks for TB, including tobacco and alcohol use, homelessness, HIV, malnutrition and incarceration.^{3,4,11–14} Compared to people who do not use drugs, people who use drugs and have TB disease are more likely to delay seeking care, be smear-positive, take longer to achieve negative cultures, and are less likely to complete therapy,^{12–14} all of which facilitate TB transmission.

Although the Stop TB Partnership identified people who use drugs as a key population vulnerable to and underserved for TB in 2016,¹⁵ the relationship between smoked drug use and TB remains under-studied. Earlier studies focused on injection drug use-related risks for TB, with only a handful of studies examining TB among people who smoke drugs (PWSD).^{3,12,13} Evidence suggests chronic methamphetamine and cannabis smoking causes a wide variety of lung pathologies, which may increase susceptibility to TB infection.¹⁶ Furthermore, smoking is associated with repeated coughing, and drugs are often smoked in social groups within closed and poorly ventilated spaces,¹⁷ enhancing environmental risks for TB transmission.¹⁸

Prior studies have relied largely on data collected during routine outbreak surveillance or from retrospective patient cohorts, with limited data collected on illicit drug use.¹³ Drug use is not a routine component of TB evaluation and is subject to underreporting due to the illegal nature of the behavior and associated stigma;¹⁹ therefore, TB treatment datasets often lack data on the type, intensity, and mode of drug use. Additionally, data on TB bacterial burden (e.g., time to culture positivity) are often minimally captured at initial TB evaluation in high TB prevalence settings, hampering retrospective investigation of associations between drug use and bacterial burden.¹³ Prospectively collected information on drug use among persons initiating TB treatment is needed to inform TB control efforts, including decisions about whether to invest in community-based interventions to support earlier uptake of TB screening and treatment among people who use drugs.

The Western Cape Province, South Africa has among the highest TB incidence and smoked drug use prevalence rates nationally.^{20,21} Estimates of the prevalence of illicit drug use range between 4.5% and 9.9% of the Western Cape's general population,^{21,22} with cannabis, methaqualone, and methamphetamine the most frequently used drugs and smoking being the main route of drug administration.²² To address gaps in our understanding, this paper aims to describe 1) rates and intensity of smoked drug use in a cohort of patients initiating drug susceptible TB (DS-TB) therapy; and 2) associations between smoked drug use and biological markers of TB bacterial load, a surrogate for infectiousness.

METHODS

Participants

We report on baseline data from participants in the Tuberculosis Treatment and Alcohol Use Study (TRUST) between May 2017 and March 2022 in Worcester, South Africa. Participants were ≥ 15 years old, had microbiologic confirmation of *Mycobacterium tuberculosis*, and were initiating outpatient treatment for active pulmonary DS-TB. Exclusion criteria included rifampicin resistance, taking epilepsy medication, receiving non-standard TB therapy, TB treatment in the prior 2 years, unknown HIV status, pregnancy, and unable/unwilling to participate in directly observed therapy.²³

Measures

Participants completed interviewer-administered questionnaires and provided blood and sputum samples and chest radiographs after TB diagnosis and within 1 week of TB treatment initiation. All data collection occurred at the facility. Information on biological sex, age, race, unemployment, education, and incarceration history were included. Body mass index (BMI) was calculated. Diabetic status was assessed using glycated hemoglobin (HbA1c) test results and prior diagnosis, categorized as normal, pre-diabetic, and diabetic. HIV status and treatment information were extracted from medical records. A study radiologist examined chest radiographs for cavitation.

Participants who self-reported methamphetamine, methaqualone, or cannabis use in the past 30 days and/or had a positive urine drug screen were classified as currently PWSD. The standard definition of current drug use is any drug use in the past 30 days.²³ Interviews and urine drug screens examined but did not detect illicit opiate use or the use of cocaine, and other amphetamine-type substances in the absence of smoked drug use. Cocaine and amphetamine-type substances were minimally detected. Drug use frequency was calculated as the number of days of any smoked drug use in the last 30 days, categorized into abstinence (0 days), occasional use (1–5 days), or frequent use (≥ 6 days). Past year drug use severity was assessed using the 11-item Drug Use Disorders Identification Test; composite scores were classified as low risk (< 8), moderate risk (≥ 8 and < 25), and severe risk of a drug use disorder (≥ 25).²⁴ Polysubstance use was defined as use of > 1 type of smoked drug. Excessive alcohol use was defined as an Alcohol Use Disorders Identification Test score ≥ 8 ,²⁵ and/or a phosphatidylethanol (PEth) result ≥ 50 ng/mL. PEth is a biomarker of past 21-day alcohol consumption.²⁶ Tobacco use was defined as any self-reported current use.

Previous TB disease was collected via self-report and from medical records. To examine bacterial burden, we collected study-specific sputum specimens. A concentrated Ziehl-Neelsen (ZN) smear was prepared and examined, followed by cultures in Mycobacterial Growth Indicator Tubes (MGITTM; BD, Franklin Lakes, NJ, USA) supplemented with polymyxin-B, amphotericin-B, nalidixic acid, trimethoprim, azlocillin antibiotics or oleic acid, bovine albumin, sodium chloride, dextrose and catalase. Growth was monitored using the BACTECTM MGITTM 960 system (BD Diagnostics Systems).²⁷ Acid-fast bacilli (AFB) presence using ZN staining and mycobacterial time to culture positivity (TTP) in days were

recorded for uncontaminated, culture-positive samples.²⁸ These analyses were performed at the project laboratory.

The South African Medical Research Council (Pretoria, South Africa), Boston Medical Center (Boston, MA, USA), University of Cape Town (Cape Town, South Africa), and Stellenbosch University (Tygerberg, South Africa) provided ethical approval for the study. All participants provided informed consent.

Statistical analyses

We used R v3.6.0 for analyses, with significance set at $P < 0.05$. We examined the frequency (%) of categorical variables and median value (interquartile range [IQR]) of continuous variables. χ^2 and Mann–Whitney U -tests were used to examine potential relationships between variables and smoked drug use. We used a proportional hazards mixture cure model for the TTP outcome. This model excluded all samples with contamination. Logistic regression was used to model smear and cavitation outcomes. All models included smoked drug use as the explanatory variable and were adjusted for sex, age, HIV status and tobacco use – variables known to influence the outcomes of interest.^{12–14} In the cavitation model, we also adjusted for previous TB.¹² We considered adjusting for other variables that were associated with smoked drug use (Table 1) and one or more outcomes (Supplementary Tables 1–3) at $P < 0.2$ (i.e., diabetic status, BMI, unemployment, and incarceration). We chose not to adjust for these variables as the direction of the relationship between these variables and smoked drug use was unknown.

RESULTS

Proportion and patterns of smoked drug use

Of the 303 participants enrolled, 88.4% ($n = 268$) used alcohol excessively and/or smoked drugs. More than half the participants smoked drugs ($n = 166$, 54.8%; Table 1). The median age of smoked drug use initiation was 16 years (IQR 14–17). Cannabis was the most frequently used drug ($n = 151$, 49.8%), followed by methaqualone ($n = 125$, 41.3%) and methamphetamine ($n = 91$, 30.0%). Among PWSD, polysubstance use ($n = 119$, 71.7%) and tobacco use ($n = 125$, 75.3%) were common. There was limited discrepancy between self-reported drug use and urine drug screen results: 35 of the 303 participants (11.6%) had a positive urine drug screen for a drug they reported not using, most frequently for methaqualone ($n = 24$, 68.6%).

Of the 166 participants classified as PWSD on biological or self-reported data, 81 (48.8%) reported frequent use, 50 (30.1%) reported occasional use, and 35 (21.1%) reported no use in the past 30 days (but had a positive drug screen). Of these participants, 41 (24.7%) and 78 (47.0%) were at severe and moderate risk of a drug use disorder, respectively.

In bivariate analyses, PWSD were more likely to be male (odds ratio [OR] 3.45, 95% confidence interval [CI] 2.14–5.62; $P < 0.001$), less likely to be older (OR 0.97, 95% CI 0.95–0.99; $P = 0.001$), unemployed (OR 1.70, 95% CI 1.06–2.75; $p = 0.029$) and have a history of incarceration (OR 3.22; 95% CI 1.90–5.60; $P < 0.001$) than participants who did not smoke drugs. PWSD were more likely to be underweight (OR 2.06, 95% CI 1.25–3.4;

$P = 0.005$) than of normal weight. PWSD were less likely to drink alcohol excessively (OR 0.36, 95% CI 0.22–0.58; $P = 0.001$), but more likely to use tobacco (OR 2.17, 95% CI 1.34–3.56; $P = 0.002$).

Associations between smoked drug use and outcomes

TTP was significantly shorter in PWSD than in people who did not smoke drugs in unadjusted analyses (hazard ratio [HR] 1.35, 95% CI 1.04–1.76; $P = 0.009$). After adjusting for age, sex, HIV status and tobacco use (Table 2), TTP remained significantly shorter for PWSD (median 7 days, IQR 5–10 vs. median 9 days, IQR 6–13; adjusted HR 1.48, 95% CI 1.10–1.97; $P = 0.008$). The Figure presents the Kaplan–Meier curve.

In a univariate logistic regression model of smear positivity and smoked drug use, PWSD had greater odds of smear positivity than people who did not smoke drugs (OR 2.33, 95% CI 1.33–4.12; $P = 0.003$). The association between smoked drug use and smear positivity remained significant in a multivariate model that adjusted for age, sex, HIV status, and tobacco use (OR 2.28, 95% CI 1.22–4.34; $P = 0.011$; Table 3).

Smoked drug use was not significantly associated with lung cavitation in a univariate logistic regression model (OR 1.09, 95% CI 0.67–1.76; $P = 0.730$) or in a multivariate model that adjusted for age, sex, HIV status, tobacco, and previous TB (OR 1.08, 95% CI 0.62–1.87; $P = 0.799$; Table 4). In this model, previous TB was associated with greater odds of lung cavitation (OR 1.74, 95% CI 1.01–3.05; $P = 0.047$). The odds of lung cavitation were lower for participants with HIV (OR 0.29, 95% CI 0.17–0.52; $P < 0.001$).

DISCUSSION

Information on associations between smoked drug use and TB bacterial burden can inform TB control efforts,¹³ including whether PWSD require interventions to facilitate earlier TB screening and treatment. These data remain scant globally, particularly for high TB prevalence countries such as South Africa.

In our research, we found that a high proportion of patients initiating DS-TB treatment at a South African primary care clinic smoked drugs. While illicit drug use prevalence estimates range between 4.5% and 9.9% of the general population in the Western Cape,^{21,22} 54.8% of our cohort had recently smoked drugs. Under-reporting of smoked drug use is common in general population surveys due to stigma and the illegality of drug use.^{29,30} Our use of urine drug screens to verify self-reported drug use may have increased accuracy of reporting, with only 11.6% of participants underreporting use. While the addition of biological drug screens may partially account for these differences in prevalence estimates, the gap remains compellingly large.

These findings suggest that PWSD may be at higher risk for TB disease than the general population, even in a community with high TB prevalence. The WHO advocates for active TB case-finding for key populations.³¹ Based on our findings, PWSD should be considered such a group in South Africa and prioritized for community-based TB screening and prevention. We recommend that other high TB prevalence countries also use

standardized questionnaires and biological screens to thoroughly assess individuals initiating TB treatment for illicit drug use so we can quantify the need to target this key population for TB screening and that we do not miss opportunities for co-managing illicit drug use and TB.

The patterns and correlates of smoked drug use observed in this study hold implications for TB therapy completion if not recognized and addressed. Almost three-quarters (71.7%) of PWSD in this cohort were at moderate or severe risk of a drug use disorder, with 48.8% reporting frequent use in the past month and 71.7% using multiple substances. Although not directly examined here, previous studies have described associations between illicit drug use, suboptimal treatment adherence and disengagement from care.^{12,13} Given these findings, TB treatment programs should consider routinely screening people initiating TB treatment for smoked drug use. This would create opportunities for TB services to provide PWSD with tailored adherence support. Efficacious behavioral interventions to reduce illicit drug use and enhance medication adherence have been implemented in South Africa's HIV services^{32,33} and for hospital-based multidrug-resistant TB services.³⁴ These could be adapted for DS-TB services provided through primary care clinics.

Notably, we found associations between smoked drug use and smear positivity (a proxy measure of infectiousness).³⁵ This confirms previous findings of associations between smoked drug use and smear positivity.¹³ Together with shorter TTP, these findings suggest higher bacterial burden among PWSD, possibly due to delayed diagnosis or differences in clinical progression. Smoked drug use was not associated with presence of cavitory disease, a factor associated with greater lung destruction and poor treatment response.³⁵ In keeping with prior literature,¹² having a previous TB episode was predictive of cavitory disease. Almost 40% of our cohort had TB previously, with rates of prior TB equally distributed among PWSD and those who did not. Lung destruction and scarring from these prior episodes is likely to have driven cavities seen in the current disease episode, dominating any impact of the smoked drug use itself.

Some study limitations are worth noting. Due to the sample size, the study lacked power to investigate how different patterns of drug use were related to TB disease presentation. While we adjusted for tobacco use in our analyses, we were unable to disentangle the effects of tobacco and smoked drug use on TB presentation, given the high rates of co-use in this cohort. Our study sample was limited to individuals who met eligibility criteria and agreed to participate in the Tuberculosis Treatment and Alcohol Use study, which required them to engage in directly observed therapy on weekdays. This requirement may have limited participation by those with steadier employment. While the cohort is from a single geographic area, our findings have implications for other high TB prevalence communities where smoked drug use is prevalent.

This study highlights the interrelationship between smoked drug use and TB, demonstrating higher bacterial burden, leading potentially to elevated risk of onward TB transmission and poorer treatment outcomes among PWSD. South Africa's current TB strategy, like most countries globally, does not address illicit drug use, representing a missed opportunity for TB case-finding and for supporting a population vulnerable for infection and poor treatment outcomes. Active TB case-finding among PWSD may facilitate earlier detection and TB

treatment. If combined with tailored, evidence-based interventions to support reductions in drug use and improved TB medication adherence, these strategies may help interrupt TB transmission and improve outcomes in this vulnerable population and within the community at large.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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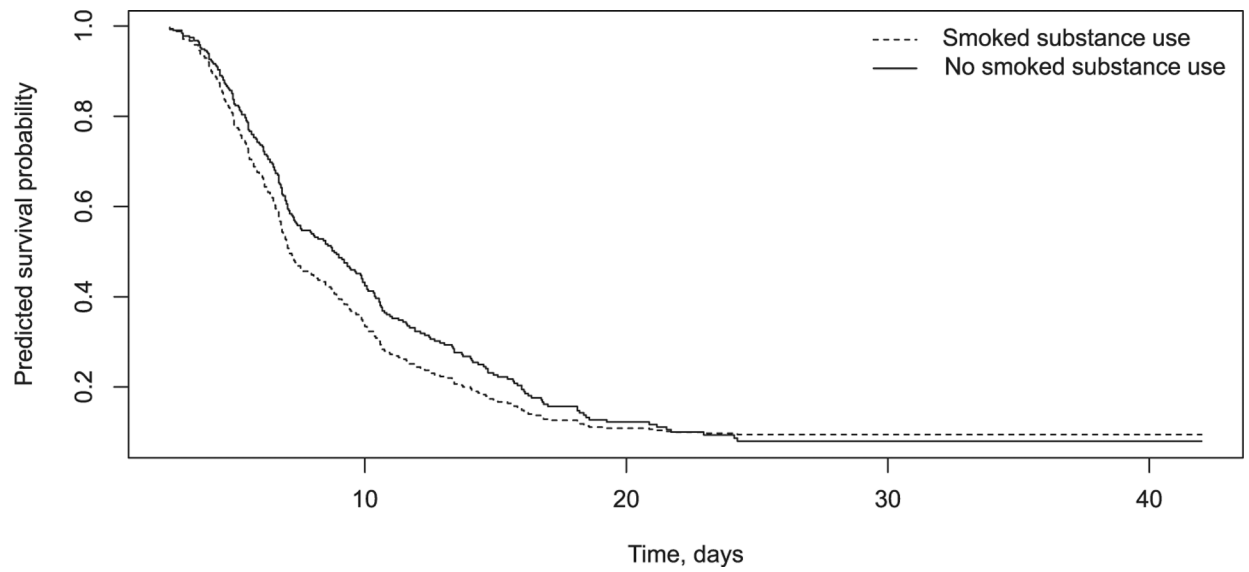


Figure.
Predicted survival curves for the estimated cure model of smoked drug use and baseline sputum culture time to positivity.

Table 1

Baseline demographics and clinical characterization for the overall sample of TRUST participants by smoked drug use status* (*n* = 303)

	Total sample (<i>n</i> = 303) <i>n</i> (%)	No smoked drug use (<i>n</i> = 137) <i>n</i> (%)	Smoked drug use (<i>n</i> = 166) <i>n</i> (%)	<i>P</i> value
Demographics				
Male	181 (59.7)	60 (43.8)	121 (72.9)	<0.001
Age, years				<0.028
<30	88 (29.0)	30 (21.9)	58 (34.9)	
30–39	78 (25.7)	34 (24.8)	44 (26.5)	
40–49	73 (24.1)	36 (26.3)	37 (22.3)	
>49	64 (21.1)	37 (27.0)	27 (16.3)	
Coloured (mixed race ancestry)	284 (93.7)	126 (92.0)	158 (95.2)	0.251
Education less than 9 th grade	129 (42.6)	54 (39.4)	75 (45.2)	0.312
Unemployed	197 (65.0)	80 (58.4)	117 (70.5)	0.028
Any history of incarceration (<i>n</i> = 302)	91 (30.1)	24 (17.5)	67 (40.6)	<0.001
Health status				
Body mass index [†]				< 0.001
Underweight	192 (63.4)	71 (51.8)	121 (72.9)	
Normal weight	95 (31.4)	52 (38.0)	43 (25.9)	
Overweight and obese	16 (5.3)	14 (10.2)	2 (1.2)	
HIV-positive (<i>n</i> = 302)	84 (27.8)	34 (24.8)	50 (30.3)	0.290
CD4 count, median [IQR] (<i>n</i> = 81)	250 [142–525]	269 [109–551]	244 [151–521]	0.617
On ART (<i>n</i> = 84)	28 (33.3)	12 (35.3)	16 (32.0)	0.753
Diabetic status [‡] (<i>n</i> = 301)				0.050
Normal	129 (42.9)	59 (43.4)	70 (42.4)	
Pre-diabetes	141 (46.8)	57 (41.9)	84 (50.9)	
Diabetes	31 (10.3)	20 (14.7)	11 (6.7)	
TB-related variables				
Previous history of TB disease	113 (37.3)	48 (35.0)	65 (39.2)	0.460
Sputum smear positivity [§]	238 (78.5)	97 (70.8)	141 (84.9)	0.003
Lung cavitation [¶] (<i>n</i> = 241)	191 (65.0)	85 (63.9)	106 (65.8)	0.730

	Total sample (n = 303) n (%)	No smoked drug use (n = 137) n (%)	Smoked drug use (n = 166) n (%)	P value
Sputum culture TTP [#] (n = 240)	170 (131, 261)	162 (121, 238)	212 (138, 303)	0.005
MGIT culture (n = 301)				0.096
Positive ^{**}	240 (79.7)	104 (77.0)	136 (81.9)	
Positive with contamination ^{††}	34 (11.3)	18 (13.3)	16 (9.6)	
Negative	23 (7.6)	9 (6.7)	14 (8.4)	
Negative with contamination	4 (1.3)	4 (3.0)	0 (0.0)	
Alcohol, tobacco and drug use variables				
Excessive alcohol use ^{‡‡}	187 (61.7)	102 (74.5)	85 (51.2)	<0.001
Tobacco use	205 (67.7)	80 (58.4)	125 (75.3)	0.002
Cannabis use ^{§§}	151 (49.8)			
Methamphetamine use ^{§§§}	91 (30.0)			
Methaqualone use ^{§§§}	125 (41.3)			
Any smoked drug use	166 (54.8)			
Polysubstance use among PWSD (n = 166)	119 (71.7)			
Smoked drug use and/or excessive alcohol use	268 (88.4)			

* Self-reported use or positive urine drug test for any of cannabis, methamphetamine, and/or methaqualone.

[†] Per NHLBI categories: underweight < 18.5 kg/m², normal 18.5–24.9 kg/m², overweight 25–29.9 kg/m², obese > 30 kg/m².

[‡] Diabetic status by HbA1c result: normal ,5.7%, pre-diabetes 5.7–6.4%, diabetes < 6.4% or history of diagnosis.

[§] AFB scanty smear-positive or higher for study or clinic sample.

[¶] Assessed using chest radiographs.

[#] Contaminated results were excluded. TTP for samples with negative growth were not included in TTP estimations.

^{**} Defined as confirmed presence of *Mycobacterium tuberculosis*.

^{††} Contamination defined as presence of contamination on Ziehl-Neelsen smear or in mycobacterial growth indicator tubes or blood agar.

^{‡‡} Participants with an AUDIT score of ≥8 or a dried blood spot PEth value >49 ng/mL.

^{§§} Self-reported use and/or positive urine drug test.

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TRUST = Tuberculosis Treatment and Alcohol Use Study; IQR = interquartile range; ART antiretroviral therapy; TTP = time to positivity; MGIT = Mycobacteria Growth Indicator Tube; PWSD = people who smoked drugs; NHLBI = National Heart, Lung, and Blood Institute; HbA1c = glycated hemoglobin; AFB = acid-fast bacilli; AUDIT = Alcohol Use Disorders Identification Test; PEth = phosphatidylethanol

Table 2Cure model of smoked drug use and baseline sputum culture time to positivity ($n = 239$)^{*†}

	aHR	95% CI	P value
Age, years [‡]			
30–39	1.23	0.82–1.84	0.311
40–49	1.11	0.74–1.65	0.617
>49	0.94	0.65–1.38	0.766
HIV status (positive)	0.46	0.32–0.65	<0.001
Biological sex (male)	1.01	0.77–1.34	0.918
Tobacco use (yes)	1.19	0.88–1.61	0.248
Smoked drug use [§]	1.48	1.10–1.97	0.008

* Model adjusted for age, sex, HIV, tobacco use.

† Contaminated samples were excluded, leaving 240 samples for analysis, 1 sample excluded from adjusted models due to missing HIV status.

‡ Reference group: < 30 years.

§ Reference group: no smoked drug use.

aHR = adjusted hazard ratio; CI = confidence interval.

Table 3Logistic regression model of smoked drug use and baseline AFB smear positivity ($n = 302$)^{*}

	aOR	95% CI	P value
Age, years [†]			
30–39	0.90	0.39–2.06	0.794
40–49	0.68	0.30–1.53	0.352
>49	0.92	0.38–2.27	0.859
HIV status (positive)	0.32	0.17–0.59	<0.001
Biological sex (male)	1.32	0.71–2.45	0.377
Tobacco use (yes)	1.26	0.68–2.29	0.461
Smoked drug use [‡]	2.28	1.22–4.34	0.011

^{*} Logistic regression model adjusted for age, sex, HIV, tobacco use.

[†] Reference group: , < 30 years.

[‡] Reference group: no smoked drug use.

AFB = acid-fast bacilli; aOR = adjusted odds ratio; CI = confidence interval.

Table 4Logistic regression model of smoked drug use and lung cavitation ($n = 293$)^{*,†}

	aOR	95% CI	P value
Age, years [‡]			
30–39	0.62	0.30–1.28	0.197
40–49	0.69	0.33–1.44	0.316
>49	0.40	0.18–0.86	0.316
HIV status (positive)	0.29	0.17–0.52	<0.001
Biological sex (male)	0.86	0.49–1.50	0.584
Tobacco use (yes)	0.95	0.54–1.65	0.857
Previous TB (yes)	1.74	1.01–3.05	0.047
Smoked drug use [§]	1.08	0.62–1.87	0.799

* Logistic regression model adjusted for age, sex, HIV, tobacco use.

† Nine participants were missing chest X-rays and were excluded from the unadjusted analyses, 1 additional participant was excluded from the adjusted model due to missing HIV status.

‡ Reference group: < 30 years.

§ Reference group: no smoked drug use.